

Sketching the future: workshop on advanced data sketching tools

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Abstract

Sketches are approximation techniques that summarize massive amounts of data into compact data structures. By operating directly on sketches, approximate analytics can be performed orders of magnitude more efficiently than exact computations on full datasets. This efficiency stems from the fact that sketches are typically tailored to specific query types, such as frequency estimation or membership queries. While most existing sketching techniques focus on aggregation queries, the underlying theory has much broader potential, with applications in areas such as machine learning and privacy. The workshop aims to bring together academic researchers developing novel sketching techniques and industry practitioners facing real-world data challenges. Our goals are to bridge the gap between research and practice, identify open problems that will shape the field over the coming decade, and develop a shared vision for the future of the data sketching community through interactive discussions and collaborative sessions. The outcomes of the workshop will be summarized in a post-workshop white paper.

1 Scientific Case

Many data systems nowadays have to manage some form of streamed data, such as stock trading data, internet traffic or sensors of a physical system for digital twins. The key property of this type of data is that records are continuously generated, often by multiple sources. Users want to analyze this data in real time, with low latency. An example analytic is to track the number of messages sent by a specific IP-address on a network stream, which is useful for applications such as monitoring whether resources are spent adequately (for companies like Ziggo, AT&T), whether malicious activity is taking place in your network (for cybersecurity software) or which parameter setting has the maximum likelihood (in machine learning [6]).

The challenge in analyzing this data lies in that we cannot use the complete data stream at all times. In many cases, the size of the data is too large to store and index, or can take too long to process and thus leads to a long queue of new data that needs to be stored. Even if we would be able to store the full data, the nature of these real-time analytics requires the queries to be answered in the order of milliseconds, which is not feasible on such large scales, if the data is not indexed properly.

Over the past decades, our community has been working on a variety of solutions based on **synopses**, small-space summaries that process the stream in one pass, where records in the stream are either processed directly, or discarded. Queries are then evaluated on the synopsis rather than on the full stream, trading exactness for efficiency. Synopses always balance several competing factors: the time required to process new data (*ingestion time*), the time needed to answer queries (*query time*), the *memory consumption* of the summary and the *accuracy* of estimates. As a general rule, allocating more memory improves accuracy, but also increases both ingestion and query costs. A wide range of synopsis techniques have been proposed, ranging from simple approaches such as sampling [9] and histograms to more sophisticated ones, called sketches [1, 3, 4].

Sketches. Sketches are a type of synopsis that serve a similar purpose to samples and histograms: compressing the data so that it occupies less space, and then using the compressed representation (the sketch) to support analytics and learning. Most sketches rely on basic randomization tools – typically hashing and repetition – to allow summarization with a single pass over the data. This makes them very attractive in the streaming context, where each data point can be observed only once, but also in

the many use cases where the data volume is very large – think, for example, training of AI models. At a high level, a sketch works by applying a randomized projection S that maps a high-dimensional dataset $\mathcal{D}$ into a much smaller representation $S(\mathcal{D})$. Then, rather than computing any function $f(\mathcal{D})$ on the original data, we can approximate f by computing another function $f'(S(\mathcal{D}))$, on the sketch, i.e., the original data does not need to be stored. Sketches allow such estimates with provable error guarantees, for select functions f .

Intuitively, this means that questions about the original, large domain and full dataset can be answered by operating on a significantly smaller representation, with a bounded error. Sketches are typically tailored to specific classes of queries, making them highly efficient for those tasks, though less general than methods like sampling.

1.1 Challenges

Despite substantial theoretical progress and meaningful impact on the practice of computer science [2], many open questions remain about the broader potential of sketches [2, 5]. These typically fall in one of the following two categories:

Adoption gap While a few sketches, as the Bloom-Filter, HyperLogLog and Count-Min sketches, are widely adopted in practice, many others remain confined to the academic world. This represents a loss of potential impact. Technical approaches to lead to more impact focus mainly on generalizing the expressivity of sketches [8, 7], aiming to make them more flexible and broadly applicable. However, adoption is not determined by technical properties alone. It also depends on factors such as how results are communicated, where they are published, their timing, ease of implementation and how well their motivation matches practical needs. Understanding these technical and non-technical challenges is essential to bridging the gap between research and practice.

Novel opportunities arising from the advancements in AI At the same time, advances in data-intensive applications and AI systems highlight the need to identify novel areas and domains where sketches can play a role. The high cost of training and inference, combined with large-scale (and sometimes even distributed) data, creates strong demand for efficient summarization techniques.

Sketches are well-suited to these settings, as they compactly represent data distributions and enable approximate reasoning. This makes them promising for tasks such as detecting concept drift in continual learning or communicating model statistics in resource-constrained environments (e.g., edge devices or federated systems). More broadly, AI workflows often tolerate approximation, making sketch-based methods a natural fit. Early work has begun to explore these directions, including applications in continual learning and federated analytics, but it is necessary to identify other roles for sketches in AI systems as well.

1.2 Workshop on advanced data sketching tools

In this workshop, we will bring people together from industry and academia, to identify and prioritize open problems in the use of data sketches in modern data-intensive and AI-driven systems. We aim to bridge the gap between academic research and industry needs by aligning problem definitions with real-world constraints such as scalability, latency and distribution drift over streams.

We invite participants from three research communities:

- Sketch algorithm designers
- AI/ML researchers working on model efficiency and robust online learning.
- Systems researchers building data platforms and large-scale ML infrastructure

These communities frequently share common challenges in terms of data management, yet their assumptions and evaluation criteria often diverge. We complement the team of researchers with industry practitioners (both in AI and data) who will contribute as problem owners, providing use cases and constraints and grounding the discussions to the real world. At the same time, practitioners will shed light on the roadblocks they are facing today on adopting sketches in their day-to-day operations.

Workshop goals

Our **first goal** is to bridge the gap between industry and academia. There are tens of sketch papers published every year, but there is no good overview on why some research is actually implemented, and why some algorithms do not make it out of the academic literature. Besides teaching each other about existing problems and solutions, it would also be helpful to discuss design requirements and communication channels for sketches such that they can be adopted more easily in industry. This overview should identify the technical and non-technical roadblocks to bridge the gap between research and practice and utilize the potential of sketch tools.

Our **second goal** is to create a common vision on what the key open problems are for the next decade that can benefit from the development of existing or new sketches. We believe that the community is helped by asking ourselves which problems are worth solving for algorithm designers from academia, according to current industry needs? Also, which challenges deserve more of researchers' attention, particularly considering the recent advancements in AI and ML? Novel understanding on open problems by attendants of the workshop would be great, but we can better serve the global community by communicating the outcomes of the workshop during the conferences in our community and other events. This can also lead to a continuous process of revising the vision for the community.

Our **third goal** is to sustain existing and foster new professional collaborations among academic and industrial partners, with a focus on advancing the workshop's agenda.

In short, this workshop will be considered a success if (1) the academic partners have a better understanding on what makes their research outcomes more applicable to be used in industry, (2) the industrial and academic partners reach novel understanding on the open problems for which we need to design advanced sketching tools during the coming decade and (3) the workshop leads to more professional collaboration between the participants.

We anticipate that this initiative will launch a series of workshops, to be hosted and supported by the Lorentz Center over the following two or three years. The aim of this workshop series is to provide recurring forum for participants to review progress and outcomes, refine plans, and progressively broaden participation by including contributors from additional sectors.

1.3 Expected outcomes

The expected workshop outcomes are:

- Descriptions of (families of) industry problems requiring advanced data sketching tools.
- Understanding and documenting the roadblocks and the steps needed to achieve broader adoption in industry. This will involve: (a) identifying the non-technical roadblocks to the adoption of sketches in practice, (b) prioritizing and addressing unsupported functionality, (c) developing guidelines for practitioners, (d) creating abstractions that unify the theory of synopses, and, (e) proposing changes in teaching, and in the long run, potentially submitting recommendations to national and international organizations like the NVAO and ACM regarding the inclusion of sketches to computer science curricula.
- Requirements for software tools, e.g., a platform that aids developers in finding the right sketch for their problems.

In the weeks/months to follow:

1. A vision statement for the community that is shareable and can be updated in the future, also by people that do not participate in the workshop. This can be presented in conferences of our community as well to encourage discussion.
2. Joint projects following the workshop, to achieve the workshops intended outcomes.
3. Research papers on solutions for the industry problems defined during the workshop.
4. Adoption of these solutions from industry partners and practitioner-oriented/technology-transfer publications and technical reports (blogs, industrial perspective articles).

2 Program

We plan to hold this workshop in three parts;

- An online preparation meeting one week prior to the onsite workshop
- An onsite workshop of three days
- An online recap meeting, one month after the onsite workshop.

Before the workshop, we will ask each participant to fill out a questionnaire on their expectations and views of what open problems there are in our domain.

We will also ask three groups of industry participants to prepare use cases that will be discussed during the onsite workshop, to focus the discussions. We plan to have use cases on **data-intensive science and sketching techniques in AI, and control systems & observability**.

The data-intensive science use case will have problem owners such as CERN, SURF and Astron, i.e., organizations that generate massive, high-velocity data streams. The use case on sketching techniques for AI will bring together researchers in AI and systems as well as companies that deploy machine learning models in practice, such as Volvo. Finally, the use case for control systems & observability will involve companies such as AstaZero and Energy Gothenburg, which manage large-scale sensor networks or edge devices and rely on monitoring and control systems to detect and respond to anomalies. These use cases are not isolated and will not be treated as such during the workshop. For instance, some control systems also deploy AI models for anomaly detection. So, besides diving deep into each use case in breakout sessions, there will be plenty of plenary discussion on common insights, and the possibility to join discussions of multiple use cases.

In addition, one of the participants will give a keynote reviewing the evolution of sketching research in the past, and outlining current directions in the field.

Day 1, online: The purpose this session will be to prepare the minds of the participants for the onsite workshop, by discussing the goals, use cases and making sure they are aware who the other participants are.

Day 2-4, onsite: Onsite workshop The onsite workshop is the core part of the workshop. This will be a three-day workshop at the Lorentz Center, in which we will discuss the industry use cases and abstract to identify open challenges. The program for these days is described below in more detail.

Day 5, online: Recap This online session will be held approximately two weeks after the onsite workshop. During this, we will recap on the onsite workshop. The white paper on the vision for the community will be presented here, and open challenges will be discussed. Also, this session gives space to discuss topics that participants thought of following the workshop.

2.1 Program of the onsite workshop

Day 1: Context, Community, and Use Cases

Time	Event	Description
09:00–09:30	Welcome & Introduction	Overview of the workshop goals, structure, and expected outcomes.
09:30–10:15	Keynote: Sketches Over the Decades	Keynote reviewing the evolution of sketching techniques and their impact over the last decades.
10:15–10:45	Coffee Break	Informal discussions.
10:45–11:30	Participant Introductions	2-3-minute introductions by participants covering background, motivation, and expectations. Important for industry partners to get to know the expertise.
11:30–12:30	Panel Discussion: Community Progress	Discussion on what the community has achieved, which research reached production, what inspired follow-up work, and why some ideas did not make it. This is to set the flavor of the remaining workshop
12:30–14:00	Lunch Break	
14:00–15:30	Industry Use Cases	Presentations and Q&A on industry use cases: • Data-intensive science • Control Systems Monitoring • Sketching techniques in AI
15:30–16:00	Coffee Break	Formation of breakout groups on use cases .
16:00–17:00	Breakout Sessions: Use Cases	Group discussions focusing on specific use cases and identifying open problems.
17:00–19:00	Reception	Informal networking and continued discussions.

Day 2: From Practice to Abstraction

Time	Event	Description
09:00–09:30	Day Opening & Recap Day 1	Summary of Day 1 discussions and goals for the day.
09:30–10:30	Breakout Sessions: Solution exploration	Groups discuss potential solutions to identified problems
10:30–11:00	Coffee Break	
11:00–11:30	Plenary Discussion on solutions	Identify abstractions of the problems
11:30–12:30	Breakout Sessions: Solutions and Abstractions	Groups discuss solutions to identified problems and abstractions to other problem settings.
12:30–14:00	Lunch Break	
14:00–14:30	Midday Recap	Groups share interim findings and insights.
14:30–15:45	Breakout Presentations and Q&A	Groups present results to each other.
15:45–16:15	Coffee Break	
16:15–17:15	Plenary Discussion & Recap	Joint discussion and synthesis of the day's outcomes.
18:00–20:00	Dinner	

Day 3: Sketching the future: Vision and Future Directions

Time	Event	Description
09:00–10:00	Keynote	Forward-looking keynote setting the stage for future research directions.
10:00–10:30	Q&A and discussion	New thoughts, reflections, and open questions.
10:30–11:00	Coffee Break	
11:00–12:00	Vision Statement Discussion	Developing a shared vision for the field.
12:00–12:30	Lessons from Use Cases	What have we learned so far from industry and practice?
12:30–13:15	Dissemination Planning	How to communicate the vision and results to the wider community.
13:15–14:15	Lunch Break	
14:15–15:45	Open Research Discussion	Focused discussion on concrete research problems and open topics.
15:45–16:00	Closing Statement	Summary of outcomes and next steps.
16:00–17:30	Drinks	Informal wrap-up and networking.

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